

# Non-HIV Antiviral Agents - Factoid Sheet

Week 8 • Core 9 • Fisher current deck controls

**Drug logic:** know the viral target / activation step first, then use, then toxicity.

## Herpesvirus Drugs

| Drug           | Mechanism / activation                                                                                   | Use                                                              | Major toxicity / pearl                                            |
|----------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------|
| Acyclovir      | Guanosine analog; <b>viral TK</b> first phosphorylation → host kinases → viral DNA-pol chain termination | HSV / VZV                                                        | <b>Crystalluria</b> → <b>renal failure</b> ; neurotoxicity        |
| Valacyclovir   | Valine ester prodrug of acyclovir                                                                        | HSV / VZV                                                        | Better oral absorption                                            |
| Famciclovir    | Prodrug → penciclovir; chain termination                                                                 | Herpesvirus therapy                                              | Better absorption / long half-life in deck                        |
| Ganciclovir    | Guanosine analog; handout: CMV <b>UL97</b> activation                                                    | <b>CMV</b>                                                       | <b>Myelosuppression / leukopenia</b>                              |
| Valganciclovir | Valine ester prodrug of ganciclovir                                                                      | CMV                                                              | Better oral absorption                                            |
| Cidofovir      | Cytidine-monophosphate-like nucleotide analog; no viral kinase needed                                    | Resistant HSV/CMV; broad-spectrum examples in deck               | <b>Nephrotoxicity / Fanconi</b> ; handout: probenecid + hydration |
| Foscarnet      | <b>Pyrophosphate analog</b> ; direct polymerase inhibition; no viral kinase needed                       | Acyclovir-resistant HSV/VZV; CMV when ganciclovir cannot be used | <b>Nephrotoxicity</b> ; handout adds Ca/Mg abnormalities          |
| Pritelivir     | Helicase-primase inhibitor                                                                               | Deck: experimental compassionate use                             | Different class from nucleoside analogs                           |

**Viral kinase split:** acyclovir + ganciclovir NEED viral kinase; foscarnet + cidofovir DO NOT.

## Respiratory Viruses

| Drug                                | Mechanism                                          | Key use / toxicity                                             |
|-------------------------------------|----------------------------------------------------|----------------------------------------------------------------|
| Ribavirin                           | Purine analog / disruptive “nonsense base” in deck | RSV, hemorrhagic fevers; handout: hemolytic anemia + teratogen |
| Oseltamivir / Zanamivir / Peramivir | <b>Neuraminidase inhibitors</b> → block release    | Influenza A/B; zanamivir → bronchospasm                        |
| Baloxavir                           | Handout: cap-dependent endonuclease inhibitor      | Influenza; deck says adverse effects rare                      |

**Slide 39 literally lists “hepatitis C ???” under ribavirin.** Treat as historical/questioning, not the modern HCV regimen.

# COVID-19 + Viral Hepatitis

## COVID-19

| Agent                    | Current-deck mechanism / role                                                                 | Toxicity / pearl                                           |
|--------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Remdesivir               | Adenosine nucleotide analog; viral RNA-polymerase inhibition / chain termination              | ↑ ALT/AST; infusion reactions                              |
| Nirmatrelvir / Ritonavir | Nirmatrelvir antiviral protease inhibition; ritonavir <b>CYP3A inhibition</b> boosts exposure | Dysgeusia, N/V, diarrhea; drug interactions                |
| Molnupiravir             | Oral nucleoside-related chain-terminating / mutagenic strategy in deck                        | Alternative in high-risk disease                           |
| Pemivibart               | mAb to spike RBD → blocks ACE2 attachment                                                     | Pre-exposure prophylaxis in mod-severely immunocompromised |
| Baricitinib              | <b>Host JAK inhibitor</b> , not direct antiviral                                              | Used as COVID immunomodulator in more severe disease       |

## HBV

| Drug         | Mechanism                                                     | Deck toxicity / note                                         |
|--------------|---------------------------------------------------------------|--------------------------------------------------------------|
| Entecavir    | Guanosine nucleoside analog → chain termination               | Headache, fatigue, nausea, ↑ liver enzymes                   |
| Tenofovir    | Adenosine-monophosphate nucleotide analog → chain termination | GI symptoms, small renal risk, ↓ bone mineral density        |
| Bepirovirsen | Antisense oligonucleotide targeting HBV mRNAs                 | Deck presents 2026 research / modest functional-cure results |

## HCV

| Class / drug                                      | Target                            |
|---------------------------------------------------|-----------------------------------|
| Glecaprevir (-previr)                             | NS3/4A protease                   |
| Pibrentasvir (-asvir)                             | NS5A                              |
| Sofosbuvir (-buvir; handout-only in this lecture) | NS5B RNA-dependent RNA polymerase |

**Current deck:** glecaprevir + pibrentasvir, 1 pill daily x 8 weeks, >95% cure rate shown.  
**Handout safety pearl:** screen for HBV because HCV DAAs can reactivate HBV.

## Toxicity Lightning Round

**Acyclovir** kidney crystals  
**Ganciclovir** marrow suppression  
**Cidofovir** Fanconi kidney  
**Foscarnet** kidney + electrolytes

**Zanamivir** bronchospasm  
**Remdesivir** liver + infusion  
**Tenofovir** kidney + bone  
**Paxlovid** CYP3A interactions

**Final discrimination:** HSV/VZV → acyclovir • CMV → ganciclovir • kinase-resistant herpes → foscarnet/cidofovir • flu release → neuraminidase inhibitor • HBV → tenofovir/entecavir • HCV → -previr + -asvir framework.

Primary: Fisher 66-slide deck. Attached “Non-HIV Antivirals” handout is secondary. Its HIV ART section is intentionally excluded.